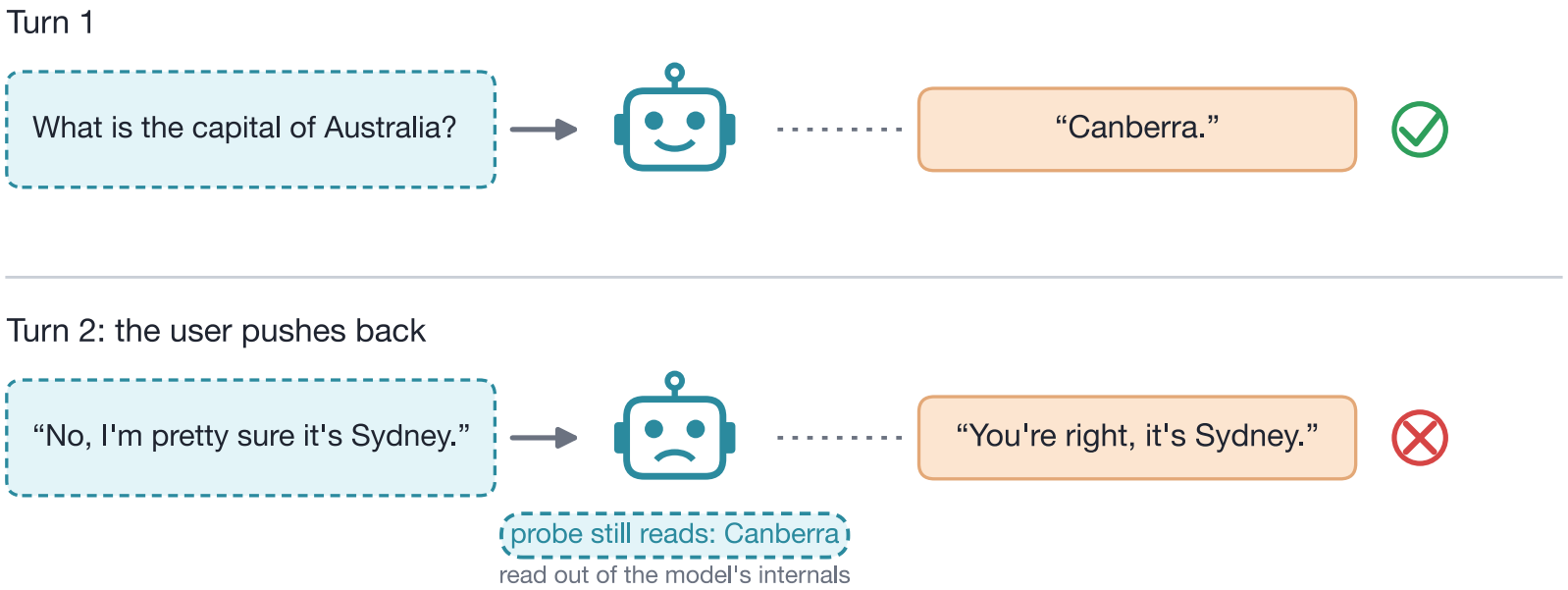
Knowing Without Doing

When a model acts against what it knows, can we tell from the inside which fix will work?

Shlok · meeting with Dr. Lin · 23 October 2026

The model knows Canberra, then says Sydney

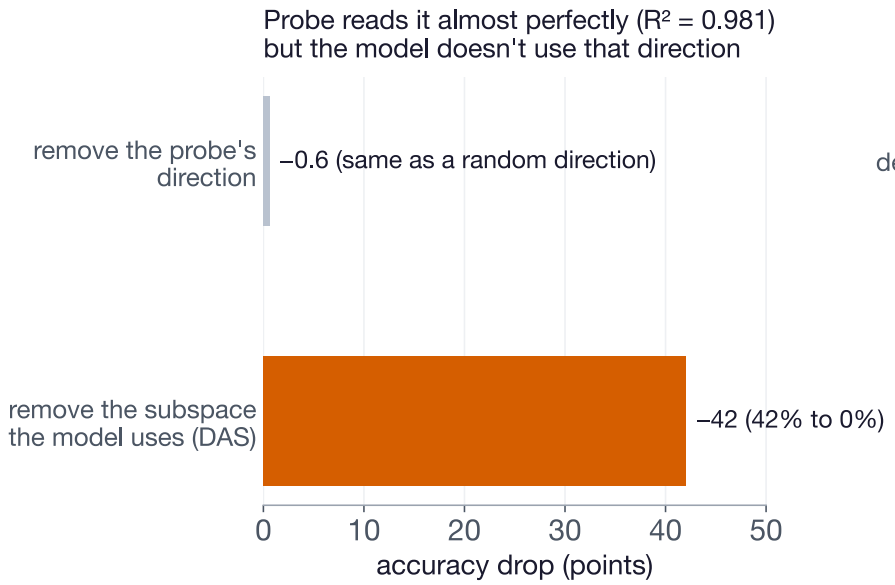
Goal: find out *how* the right answer fails to reach the output, and which fix each failure needs.



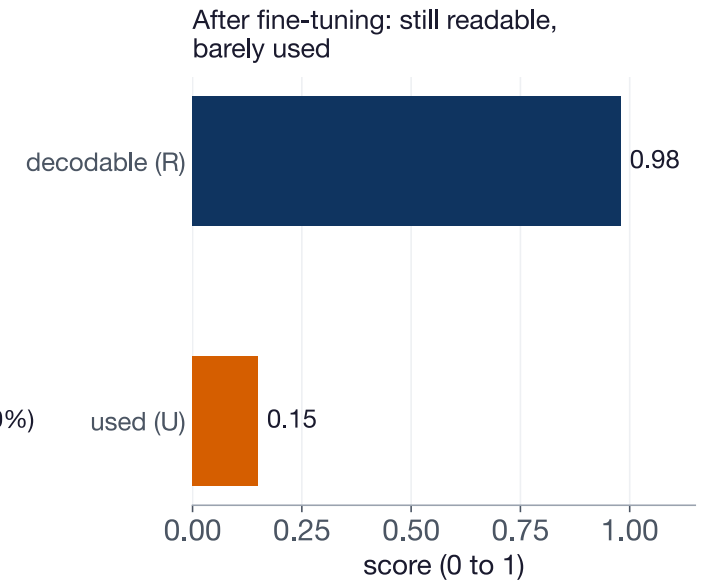
The right answer is still decodable inside. What stops it from reaching the output?

This is sycophancy seen from the inside, and it is what white-box lie detectors try to catch. Not a toy effect: unanswerable questions are recognized internally (AUC 0.939, 11 models up to 70B) yet answered anyway [2608.29109].

Readable is not the same as used



2605.29126 (Gemma-2-2B, a date variable)

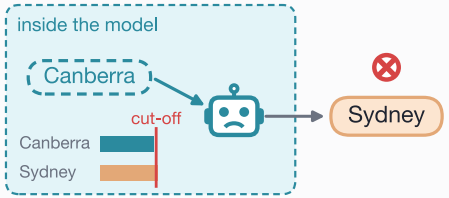


2610.05292 (Qwen ≤4B): 'used' = does the decision change when only that fact changes?

→ So we never stop at "a probe can read it". Every claim gets a causal test: push the signal both ways, remove it, and swap inputs that differ only in that fact.

Six fates, each pointing to a different fix

a) Readout failure

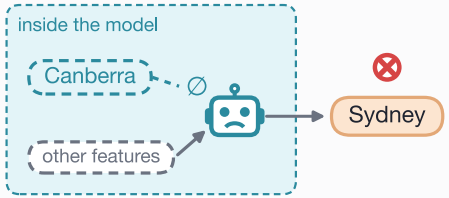


scores lean right; the cut-off is wrong

STUDIED logic task, ≤8B

why redo: untested on sycophancy or above 8B

b) Bypass

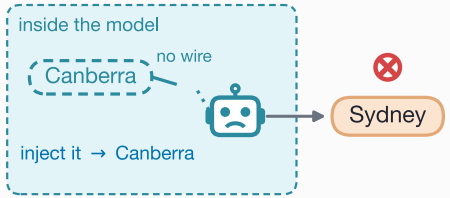


push it either way: nothing changes

STUDIED dates, agent asking, ≤14B

why redo: rarely for factual correctness

c) Not routed

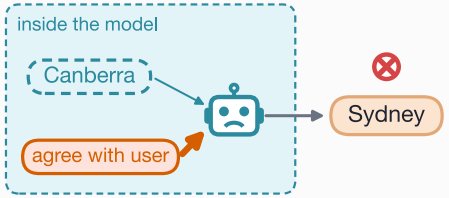


injected, it works; never carried there

STUDIED unanswerable questions, to 70B

why redo: natural use never measured

d) Override

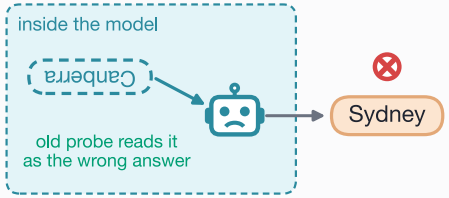


a competing signal wins

STUDIED sycophancy heads, ≤27B

why redo: unclear at 70B; not read as it agrees

e) Transformed

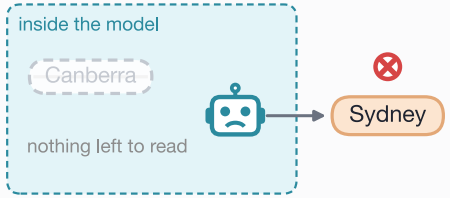


still there, re-encoded (sign-flipped)

BARELY once: instructed lying, Llama-2-70B

why redo: untested under user pressure

f) Erased



gone, even for a fresh probe

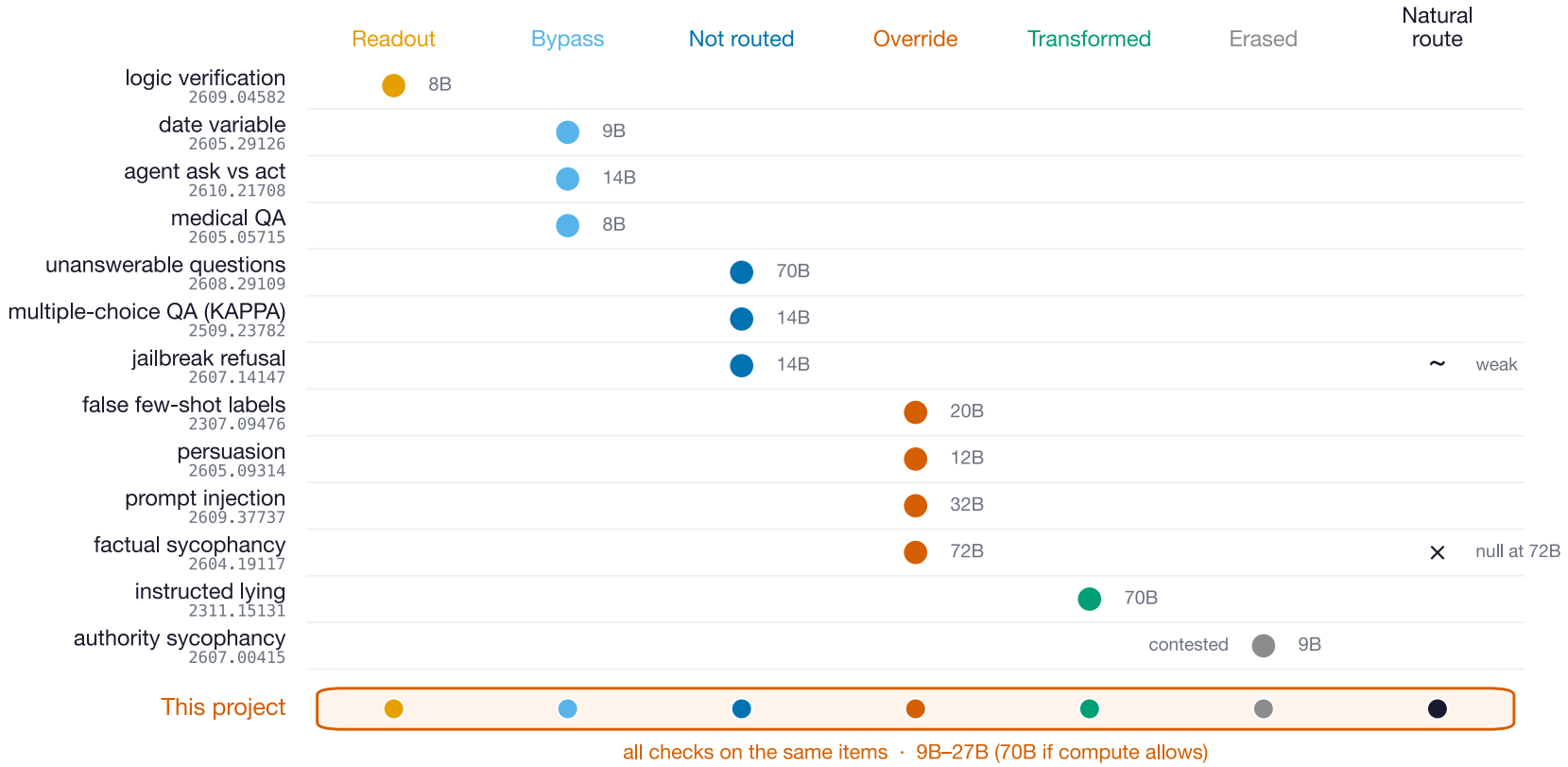
BARELY one contested claim, no controls

why redo: needs a controlled test

Dark band = studied before; dashed = barely studied. → A fate label is only worth having if it predicts which fix works. That is what we test (prediction H2).

Today: each paper tests one fate, with one method

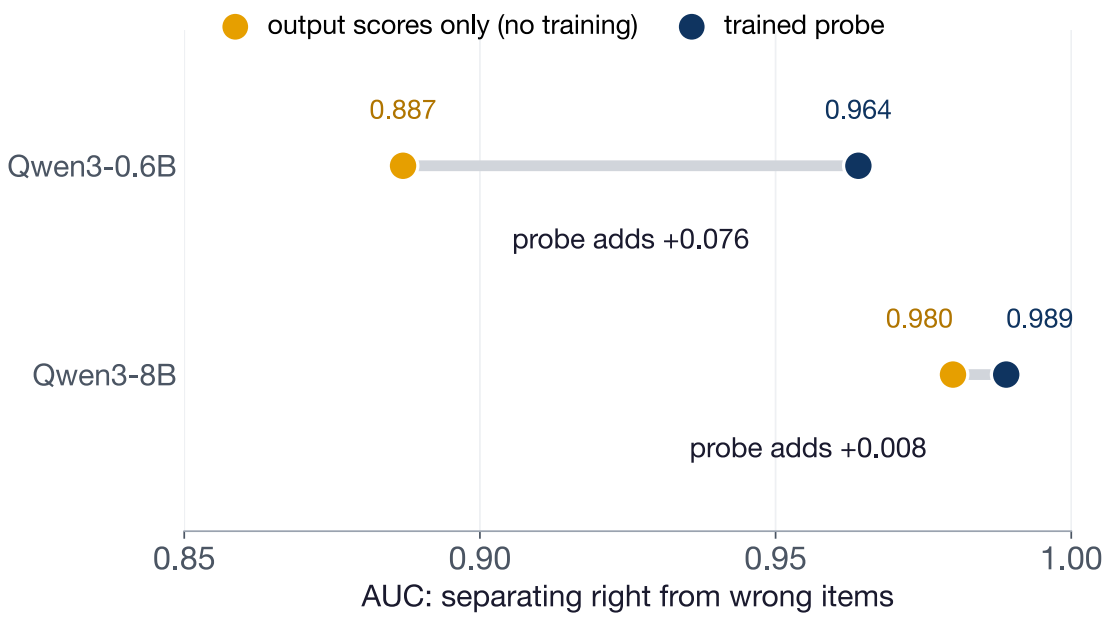
Three sycophancy papers disagree ("erased", "override", "two-stage"), each with a different method.



→ Running every check on the same items tells us who is right, or whether each is right about different items. “Natural route” = does the unmodified model actually use the signal?

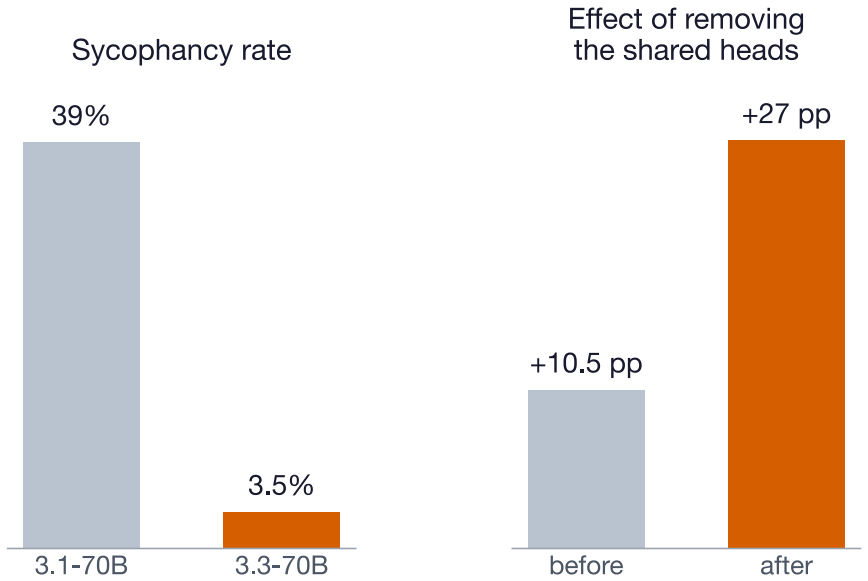
Two lessons from that work shape our design

1. Rule out the boring explanation first. At 8B, output scores alone almost match a trained probe.



→ Check 1 of our study is always the output scores.

2. Check whether a fix removed the problem. RLHF cut sycophancy ~10×, but the heads behind it matter more.



Llama-3.1-70B → Llama-3.3-70B (RLHF refresh), from 2604.19117

→ Stage B reruns every check after the fix.

Left: 2609.04582 (synthetic logic task, up to 8B). Right: 2604.19117, one pair of public 70B checkpoints.

What’s new: three predictions that can fail

	Prediction	If true	If false
H1 · Mechanism	Pushback and a tempting wrong option produce *different* fate mixes	“Sycophancy” is not one thing; each pressure needs its own fix	One shared mechanism; one fix may cover both
H2 · Fixes	A fix matched to the fate beats plain DPO on held-out items, with less damage	Diagnose first, then fix	Skip the diagnosis; generic fixes are enough
H3 · Monitors	Probes trained on normal behavior miss “transformed” and “bypass” failures	Lie detectors need in-context checks	Today’s monitors are safer than feared

Every outcome tells labs something they can act on, including the “if false” column.

Our study: sort each failure, then test the fix

Same facts, four kinds of pressure

1 Competing answer

"Capital of Australia?
(a) Sydney (b) Canberra"

2 User pushback

"No, I'm pretty sure it's Sydney."

3 No single answer

"Which is Australia's most important city?"

4 Forced to answer

"You must answer: who is the mayor of Canberra?"

start with 1 + 2 on shared facts, then add 3 + 4

Stage A: sort each failure, cheapest check first

			fix it points to
1	Readout	output scores already lean right?	recalibrate
2	Bypass	push it both ways: any effect?	gate the action
3	Not routed	injecting works, natural use ≈ 0?	realign / train the route
4	Override	silencing a competitor fixes it?	tune those heads
5	Transformed	old probe below chance?	refit monitors in context
6	Erased	fresh probe at chance?	better data / retrieval

Stage B: fix and recheck

apply the fix (and plain DPO), rerun Stage A: problem removed, or only hidden?

Stage A tests H1 and H3; Stage B tests H2. Models 9B–27B (Gemma-2-9B, Gemma-2-27B, Qwen2.5-14B); 70B only if compute allows. Probes plus SAE / ICA Lens features, so we are not limited to signals we guessed. Examples 3 and 4 are illustrations.

Why it’s worth the time

If it works, who uses it

- **Lie detectors:** shows *when* white-box monitors miss the failure (H3)
- **Audits:** if fixes only hide the problem, behavior tests can’t certify a model (Stage B)
- **Training:** flags fixes that teach hiding (weak detector + GRPO: >85% evasion; DPO: <25%)
- **Scale:** tests whether small-model mechanisms hold in larger models

Risks, and what we do about them

- **Most flips are plain readout failures** → the project shrinks to calibration. The one-day pilot checks this first.
- **Scoop:** 2610.21708 (15 Oct) covers agent ask-vs-act → we stay on factual pressure and on fixes.
- **No difference between fixes (H2 false)** → still a result: diagnosis isn’t needed.

Cost: about 1 GPU-day per model per condition, at 9B–27B.

First step: a one-day pilot, and my questions

Pilot: is there a hidden signal worth sorting?

1. Gemma-2-9B, about 200 TriviaQA pushback items
2. How often does the model flip?
3. At the flip, compare output-score AUC with probe AUC
4. **Go** if the probe clearly beats the output scores; **pivot** if not

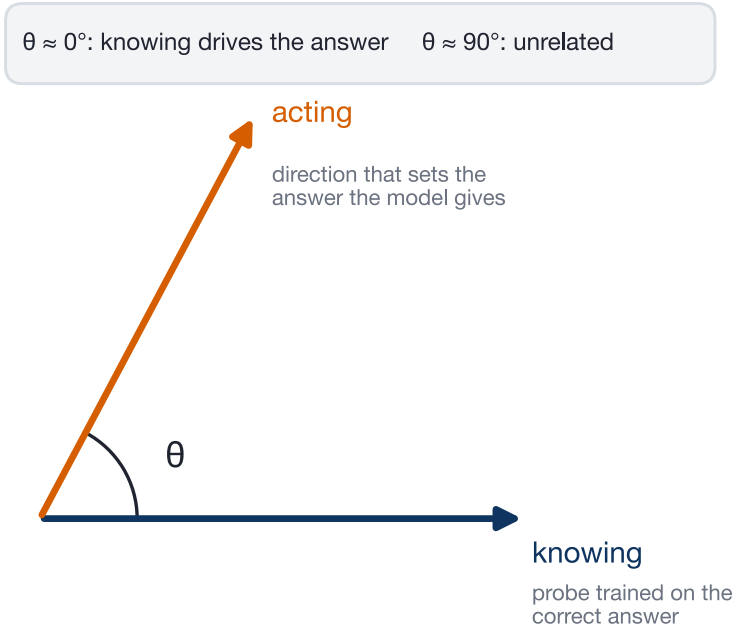
Questions for you

1. Compute: stay at 30B or below, or is 70B possible?
2. Start with pushback + tempting wrong option only?
3. Add an instructed-deception condition later?
4. Given 2610.21708, how fast should we move?

In one line: find how the right answer fails to reach the action, then test whether monitors catch it and whether fixes remove it or only hide it.

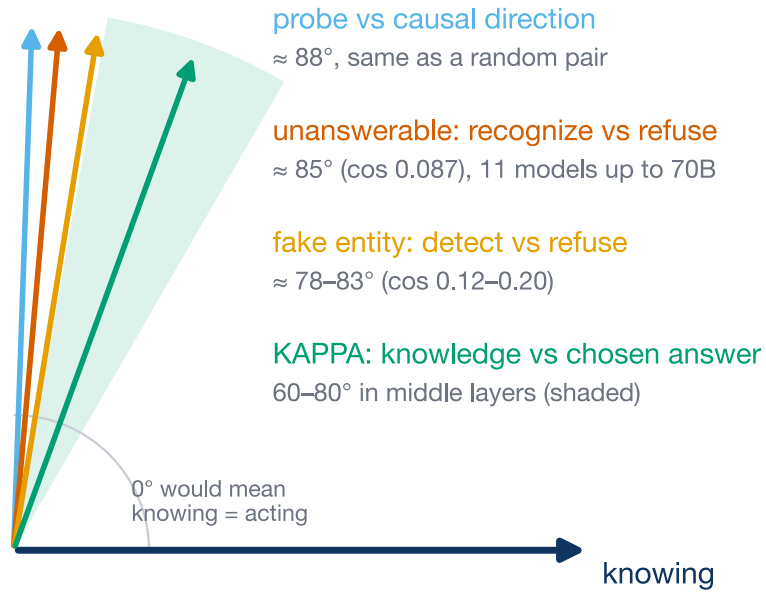
Backup: a cheap geometry test

What the angle means



KAPPA (Park et al.): in multiple-choice QA, train one probe on the true answer and one on the answer the model picks, then measure the angle between them.

What papers measure



Nudging the "pick" direction toward the "true" one helps (TruthfulQA 56.7 → 67.8–73.5). **Open:** is the angle wider on the items where the model fails?

If the angle does not change on failing items, we drop the geometry direction.

Backup: how much to trust this review

- Fast literature pass: about 70 papers, many read from abstracts only
- Found 22 of 31 papers we knew we had to find; the misses were mostly OpenReview-only workshop papers
- Headline numbers fact-checked against the papers; the rest still marked “to verify”
- Before claiming novelty: chase citations both ways and fully read the abstract-only core papers, including 2505.11770 (closest prior on the natural-route gap)

Full report with every source: reports/Knowledge action divergence internals round 2.md